

Demigration/remigration

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Modeling

Kirchhoff modeling:
Integral over reflector $z = \Sigma(x, y)$

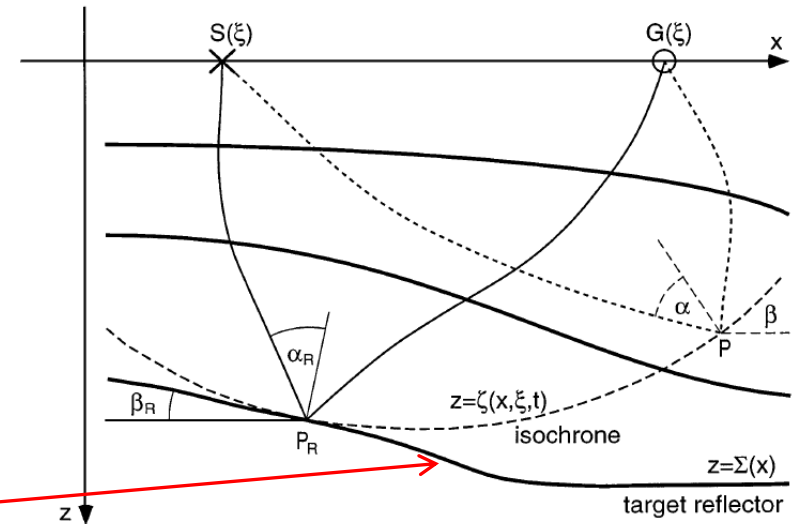


FIG. 1. Inhomogeneous earth model with smooth interfaces. Also shown is one isochron for the indicated source–receiver pair.

Forward Kirchhoff modeling: Model space $\rightarrow$ data space

$$I_K(\xi, t) = \frac{1}{2\pi} \int \int d^2x W_K(\xi, \mathbf{P}_\Sigma) \partial_t F(t - \tau(\xi, \mathbf{P}_\Sigma))|_{z=\Sigma(x, y)}$$

$$\tau(\xi, \mathbf{P}_\Sigma) = T(\mathbf{S}(\xi), \mathbf{P}_\Sigma) + T(\mathbf{R}(\xi), \mathbf{P}_\Sigma)$$

$$F(t) = \text{wavelet}$$

Migration

Kirchhoff migration:
Map data $D(\xi, t)$ to
isochrone $z = \zeta(\mathbf{x}, \xi, t)$

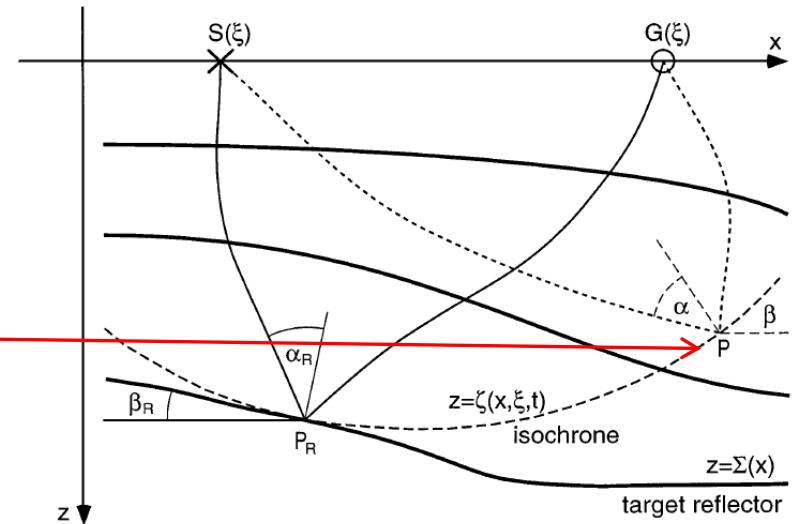


FIG. 1. Inhomogeneous earth model with smooth interfaces. Also shown is one isochron for the indicated source–receiver pair.

Kirchhoff migration: Data space $\rightarrow$ image space

$$I_M(\mathbf{P}) = -\frac{1}{2\pi} \int \int d^2\xi W_M(\xi, \mathbf{P}) \partial_t D(\xi, t)|_{t=\tau(\xi, \mathbf{P})}$$

$$\tau(\xi, \mathbf{P}) = T(\mathbf{S}(\xi), \mathbf{P}) + T(\mathbf{G}(\xi), \mathbf{P})$$

$$\mathbf{P} = \text{image point on isochrone}$$

Migration: Sum over data points $D(\xi, t)$

Demigration

Kirchhoff demigration:
Map isochrone $z = \zeta(\mathbf{x}, \xi, t)$
to data $D(\xi, t)$

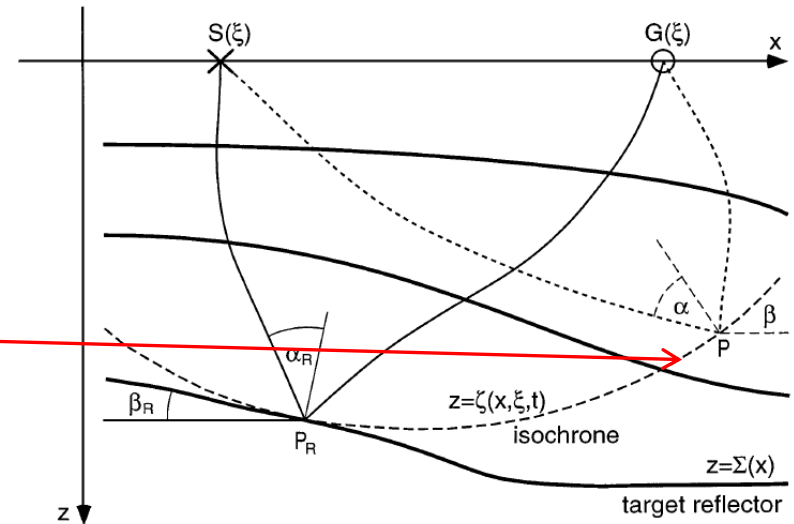


FIG. 1. Inhomogeneous earth model with smooth interfaces. Also shown is one isochron for the indicated source–receiver pair.

Kirchhoff demigration: Image space $\rightarrow$ data space

$$I_D(\xi, t) = \frac{1}{2\pi} \int \int d^2x W_D(\mathbf{x}, \xi, t) \partial_z M(x, y, z)|_{z=\zeta(\mathbf{x}, \xi, t)}$$

$$\tau(\xi, \mathbf{P}) = T(\mathbf{S}(\xi), \mathbf{P}) + T(\mathbf{G}(\xi), \mathbf{P})$$

Demigration: Sum over image points $M(x, y, x)$

Applications of demigration

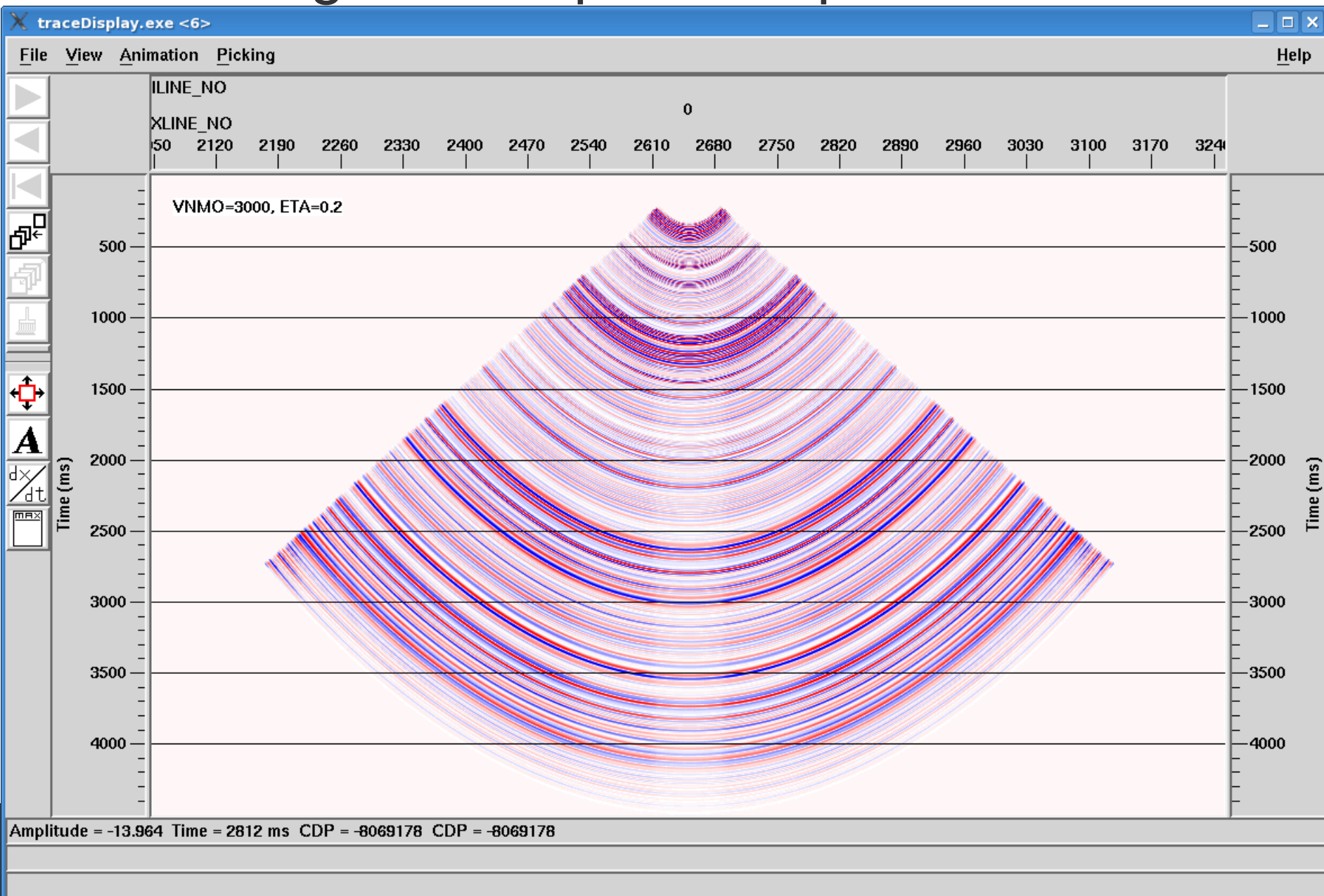
- Part of (some) tomography workflows
- Depth conversion beyond vertical stretch
 - Migration to TWT -> demigration to TWT -> remigration to TVD

Vertical-stretch depth conversion:

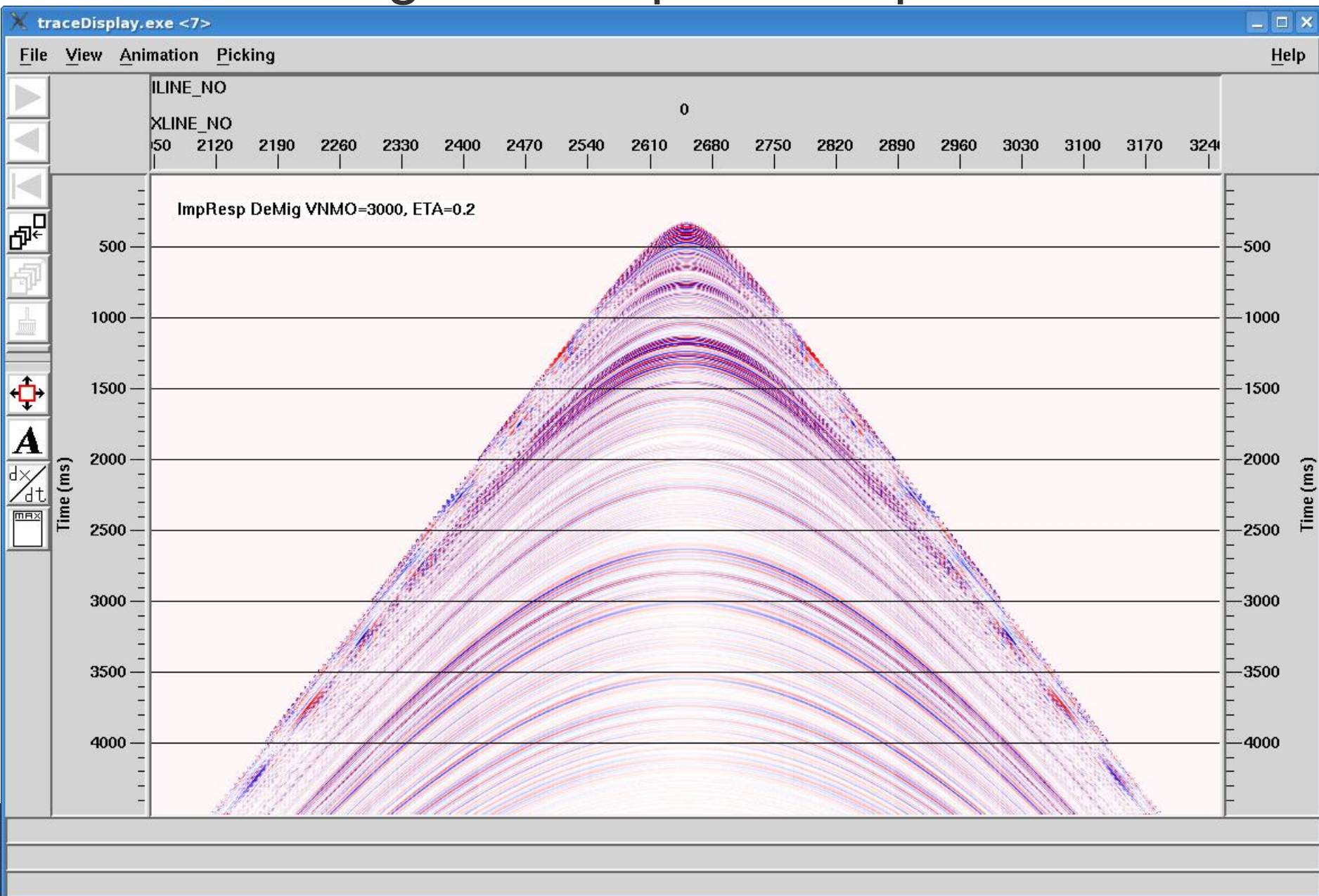
A simple process

$$z = \frac{\hat{v}t}{2}$$

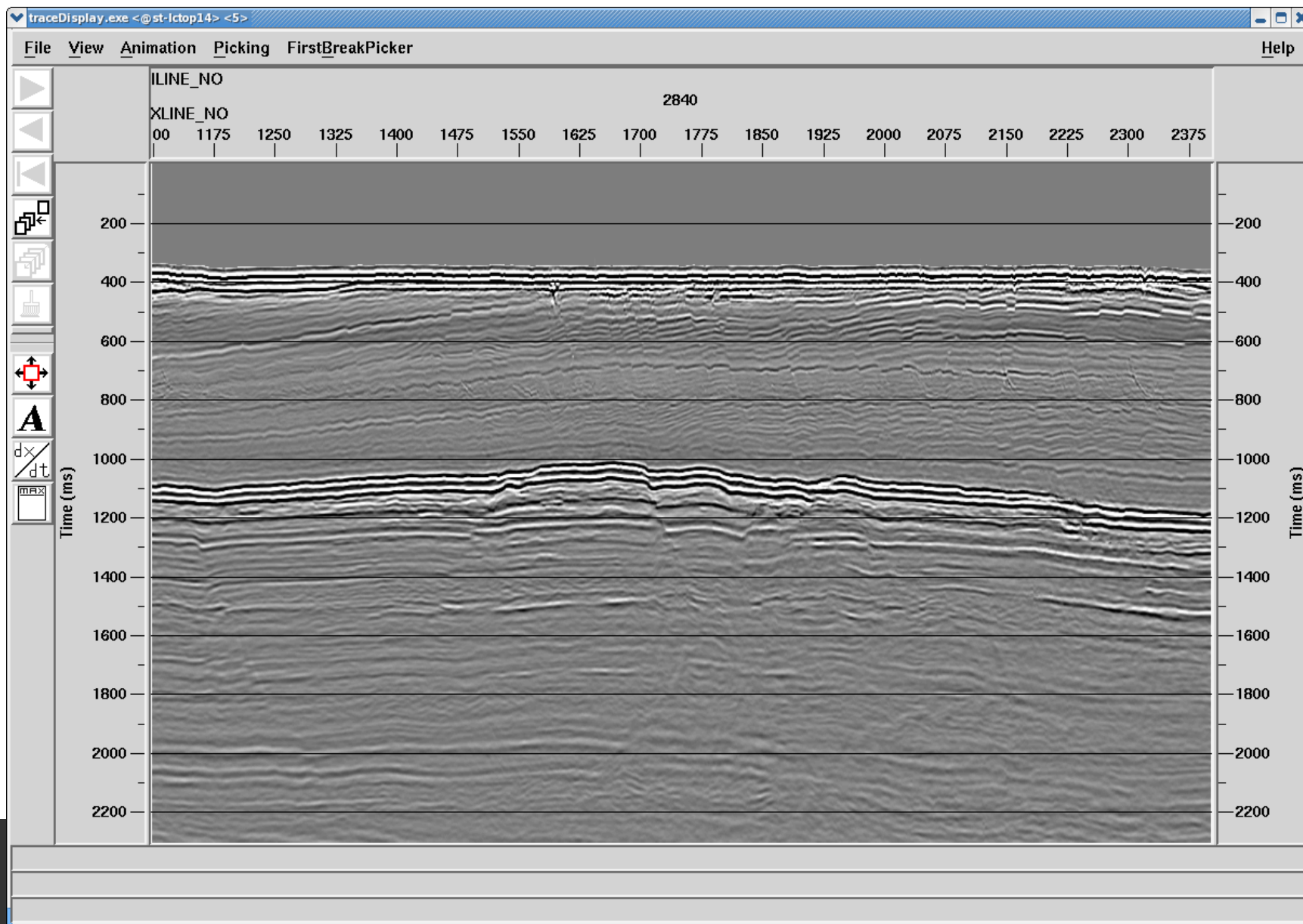
KTMIG Migration impulse response



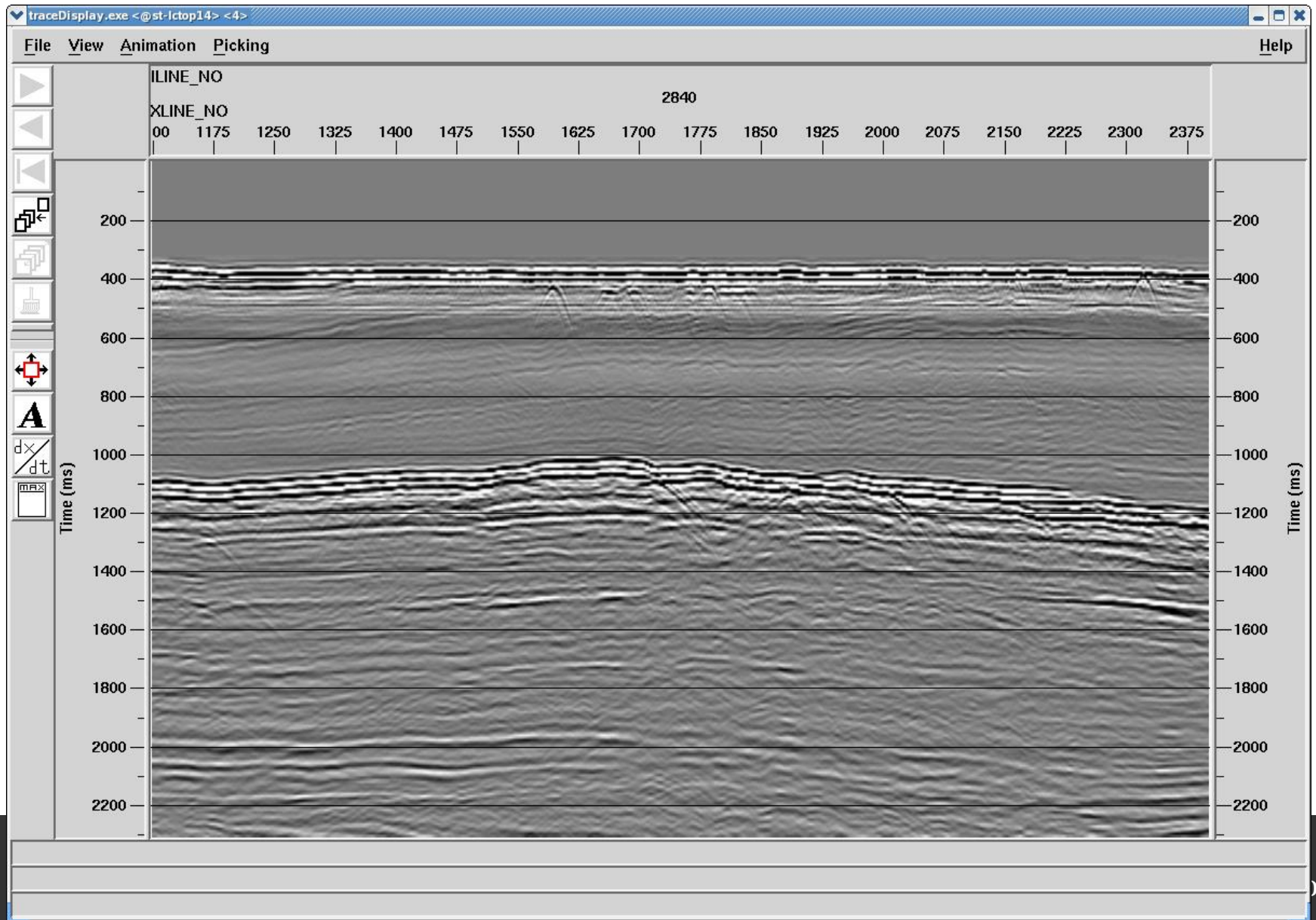
KTMIG Demigration impulse response



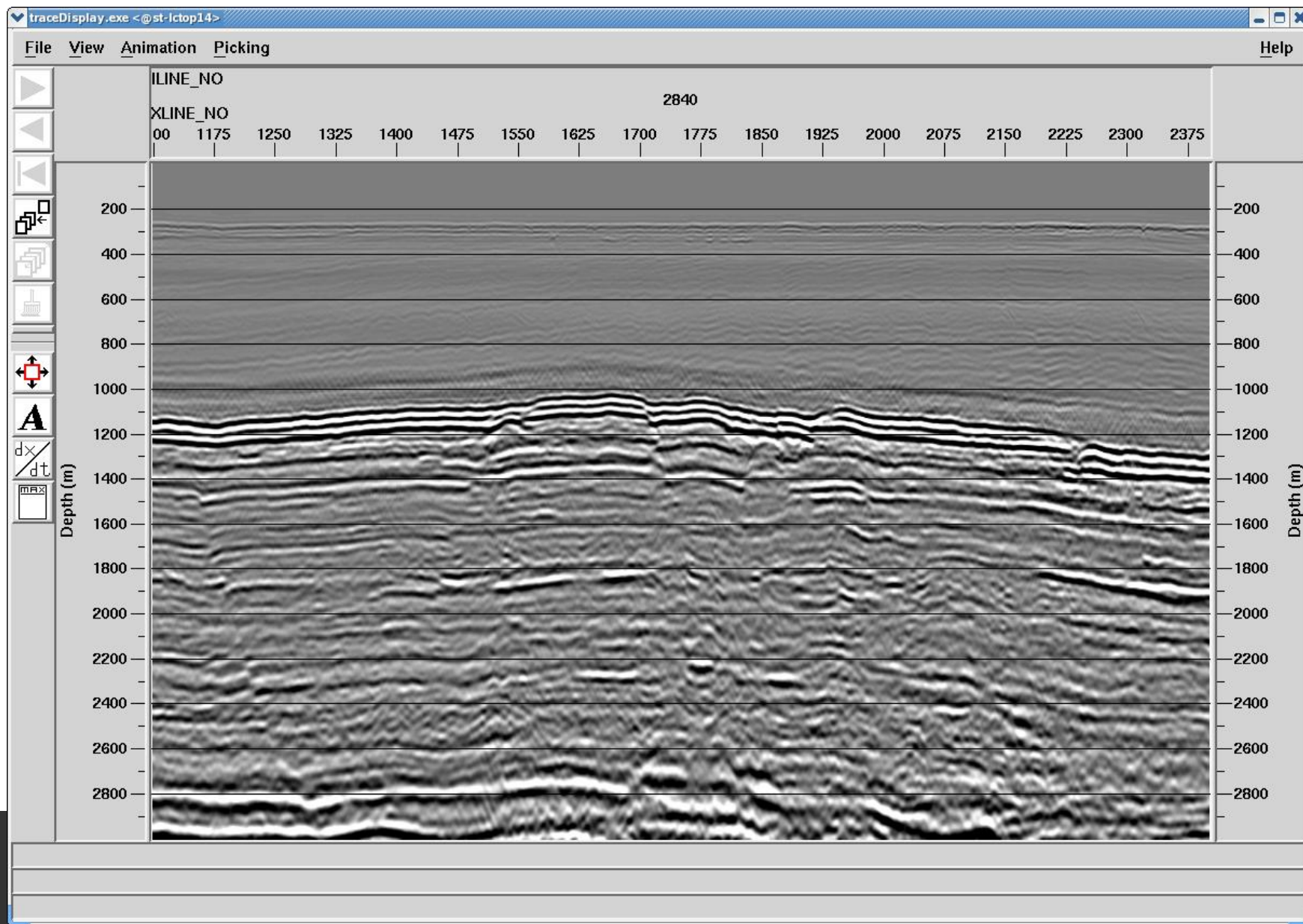
IL2840 PSTM full stack (CGG-V)



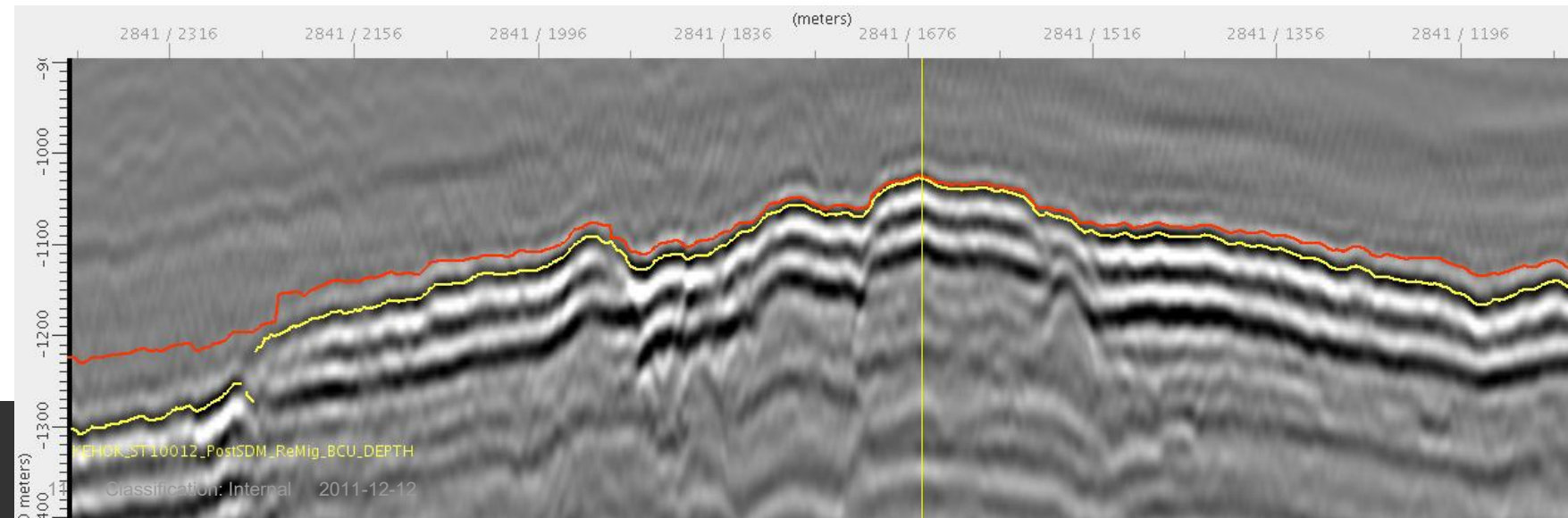
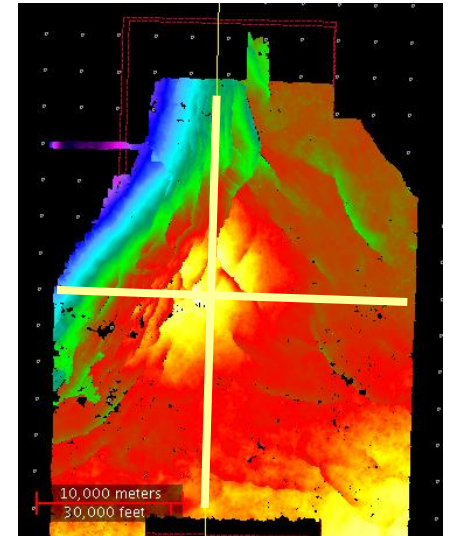
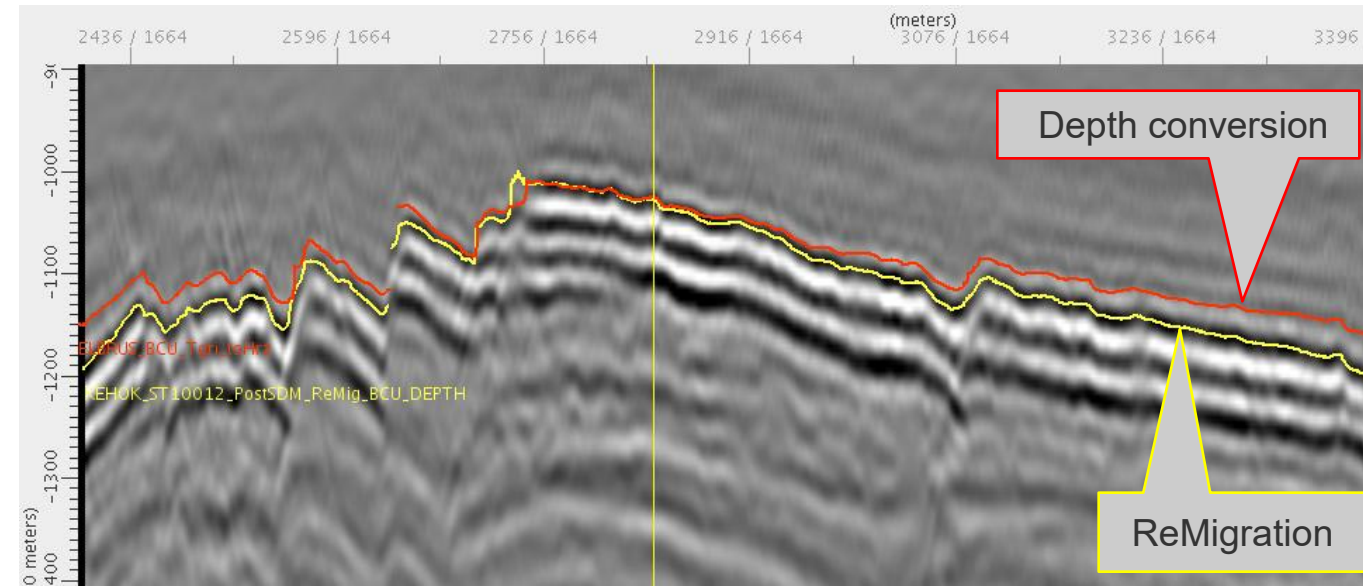
IL2840 DeMigration to TWT



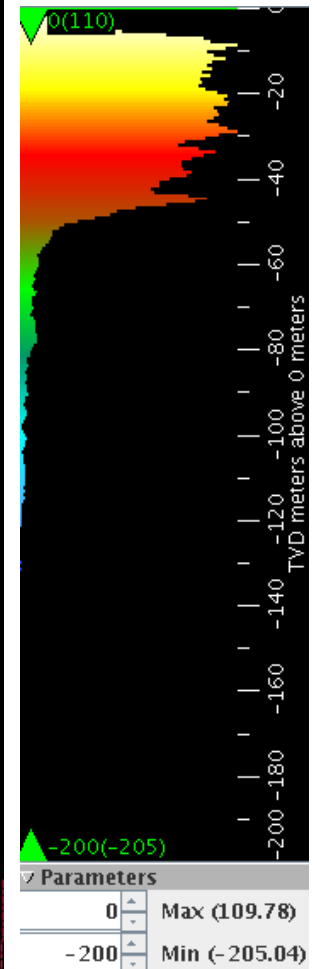
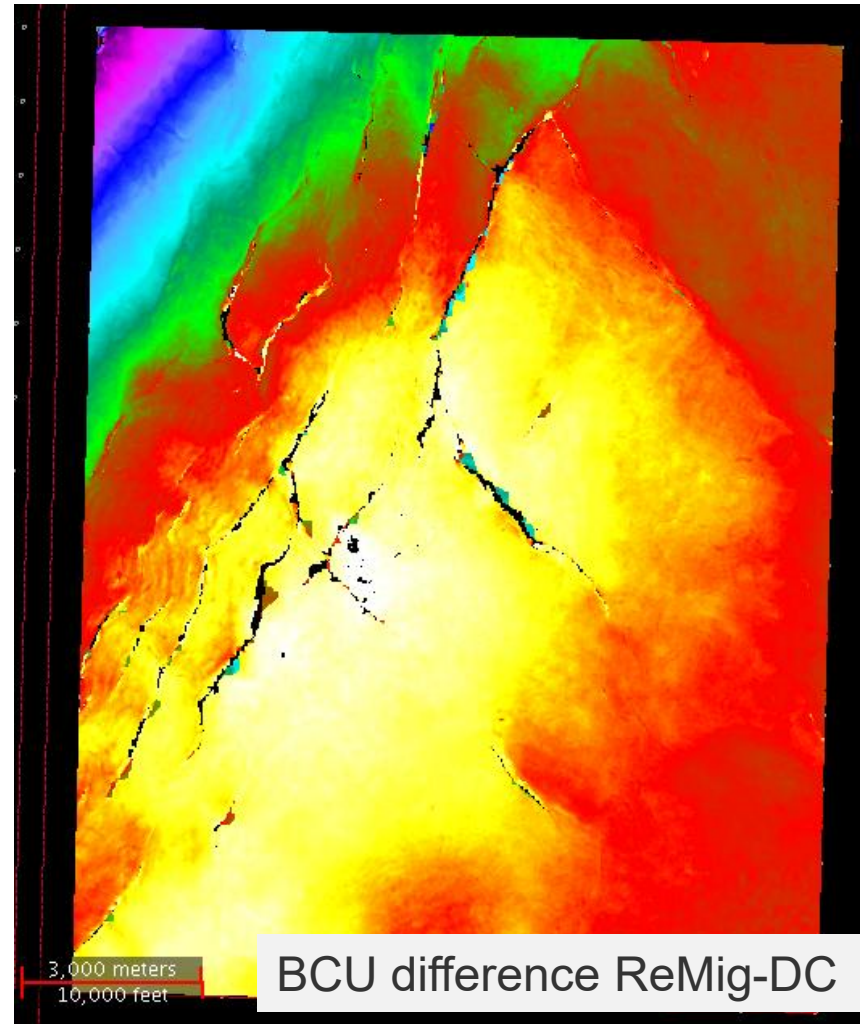
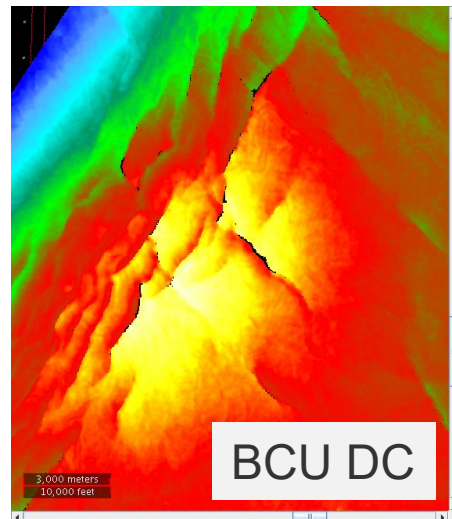
IL2840 ReMigration to depth



Vertical-stretch DC vs De/ReMigration



BCU comparison ReMig vs. vert. stretch DC



Nå kan gigantfunnet bli redusert

Skuffende letebrønn på Avaldsnes. Både operatøren Lundin og Statoil signaliserer at oljevolumene i denne delen av storfunnet vil måtte reduseres.

Espen Bjerke

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Lundin Petroleum har boret ferdig avgrensningsbrønnen 16/5-2S på det gigantiske Avaldsnes-funnet i Nordsjøen. Målet med boringen var å avgrense funnets utbredelse i sør.

Resultatet av boringen var ikke positivt for Avaldsnes. Det ventede reservoaret ble ikke funnet på dybden man antok det ville være. Den øvre delen av reservoaret ble påtruffet dypere enn forventet og befant seg under olje/-vann-kontakten, opplyser Lundin Petroleum.

